



**Year 11 Mathematics Extension 1
Assessment Task 1
Term 1 Week 4 2020**

Name:

Teacher:

Tuesday 19th May 2020

Setter: WIL

General Instructions

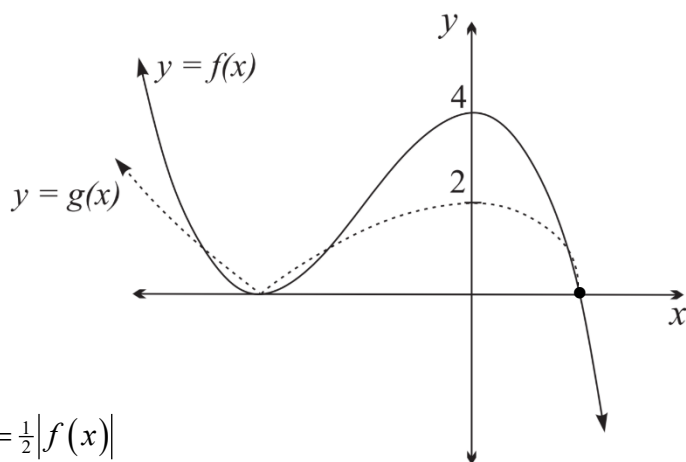
- Working time – 55 minutes, inclusive of printing or downloading, writing, scanning, naming and uploading the task to Google Classroom.
- Answer all questions on your own lined paper using black or blue pen.
- It is your responsibility to ensure that your marker can read your responses in the PDF file that you create.
- NESA approved calculators may be used and you can reference any notes and/or your textbook during the task.
- Show all relevant mathematical reasoning and/or calculations.
- **Total marks – 32.**
- Attempt Questions 1 – 4.
- All questions are of equal value.

Question 1 (8 marks)

For **Multiple Choice** questions, write down the correct option on your writing paper.

- (a) The diagram below shows the graphs of the functions $y = f(x)$ and $y = g(x)$.

Which of the following could represent the relationship between $f(x)$ and $g(x)$? **1**

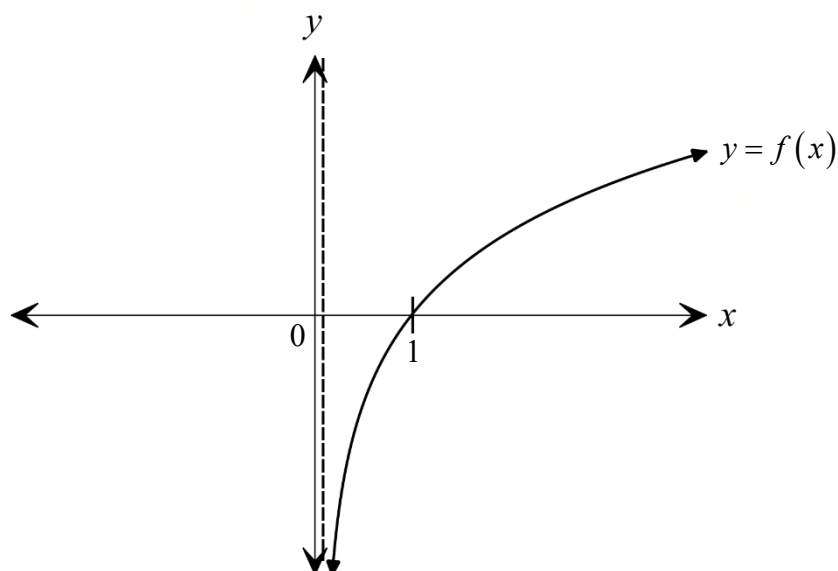


- (A) $g(x) = \frac{1}{2}|f(x)|$
- (B) $g(x) = \sqrt{f(x)}$
- (C) $g(x) = \frac{1}{f(x)}$
- (D) $(g(x))^2 = f(x)$.

Question 1 continued on page 2

Question 1 continued

(b)



Use the sheet on page 7 to sketch the following graphs, clearly showing any intercepts with axes, asymptotes and changes to the curves' shape. Please go over $f(x)$ on the template with your pen where necessary.

(i) $y = f^{-1}(x)$ 1

(ii) $y = |f(x)|$ 1

(iii) $y = f(|x|)$ 1

(iv) $y^2 = f(x)$ 1

(c) (i) Show that $\sqrt{\frac{\operatorname{cosec}^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}} = |\tan x|$. 2

(ii) Find the value(s) of x if $\sqrt{\frac{\operatorname{cosec}^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}} = 1$ for $0^\circ \leq x \leq 180^\circ$. 1

End of Question 1

Question 2 (8 marks) (Start a new page)

For **Multiple Choice** questions, write down the correct option on your writing paper.

- (a) Which of the following is a correct simplification of $\frac{\cos(180^\circ - \theta)}{\cos(90^\circ - \theta)}$? **1**
- (A) $\cos 90^\circ$
- (B) $-\tan \theta$
- (C) $\tan \theta$
- (D) $-\cot \theta$.
- (b) Solve $\cos(x - 60^\circ) = \frac{1}{\sqrt{2}}$ for $0^\circ \leq x \leq 360^\circ$. **2**
- (c) Use the expansion of $\left(\sqrt{15 - 6\sqrt{6}} - \sqrt{15 + 6\sqrt{6}}\right)^2$ to simplify $\sqrt{15 - 6\sqrt{6}} - \sqrt{15 + 6\sqrt{6}}$ leaving your answer in simplest surd form. **2**
- (d) Let $g(x) = \frac{1}{x-2}$.
- (i) Draw $g(x)$ and $\frac{1}{g(x)}$ on one diagram on your lined paper. Make your diagram at least one quarter of a page, clearly showing any intercepts with axes and asymptotes. **2**
- (ii) Let $h(x) = g(x) + \frac{1}{g(x)}$. Sketch $h(x)$ on the same diagram, clearly showing any intercepts with axes and both asymptotes. **1**

End of Question 2

Question 3 (8 marks) (Start a new page)

For **Multiple Choice** questions, write down the correct option on your writing paper.

- (a) The polynomial $P(x)$ has degree 4 and the polynomial $Q(x)$ has degree 2. If you divide $P(x)$ by $Q(x)$, the remainder $R(x) \neq 0$. The remainder has degree: **1**
- (A) 1
- (B) 2
- (C) 0 or 1
- (D) 0, 1 or 2
-
- (b) The polynomial $Q(x) = x^3 + bx^2 + cx + d$ has zeros 0, 3 and -3 .
- (i) What are the values of b , c and d ? **2**
- (ii) Without using calculus, sketch the graph of $y = Q(x)$. **1**
- (iii) Hence or otherwise, solve the inequality $\frac{x^2 - 9}{x} > 0$. **1**
-
- (c) Find the roots of the equation $x^3 + 6x^2 - x - 30 = 0$ given one of the roots of the equation is the sum of the other roots. **3**

End of Question 3

Question 4 (8 marks) (Start a new page)

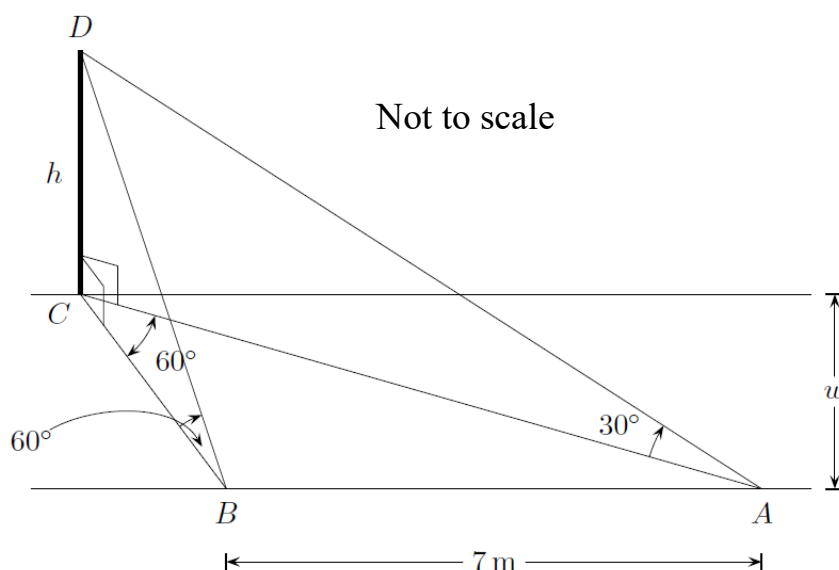
For **Multiple Choice** questions, write down the correct option on your writing paper.

- (a) It is known that $x + 2$ is a factor of the polynomial $P(x)$ and that $P(x) = (x^2 + x + 1) \times Q(x) + (2x + 3)$ for some polynomial $Q(x)$. From this information alone, it can be deduced that:

1

- (A) $Q(-2) = -\frac{1}{3}$
 (B) $Q(-2) = \frac{1}{3}$
 (C) $Q(-2) = -1$
 (D) $Q(-2) = 1$.

- (b) A footpath on horizontal ground has two parallel edges. CD is a vertical street sign pole of height h metres which stands with its base C on one edge of the footpath. A and B are two points on the other edge of the footpath such that $AB = 7$ m and $\angle ACB = 60^\circ$. From A and B , the angles of elevation to the top of the pole at D are 30° and 60° respectively.



- (i) Show that $CA = h\sqrt{3}$ and $CB = \frac{h}{\sqrt{3}}$. 1
 (ii) Show that the exact height of the flagpole is $h = \sqrt{21}$ metres. 1
 (iii) Find the area of $\triangle ABC$. 2
 (iv) Hence or otherwise, find the exact width w of the footpath. 1

Question 4 continued on page 6

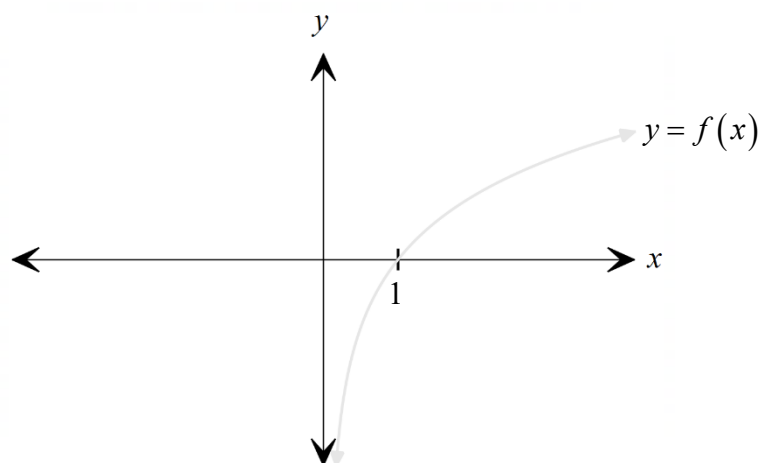
Question 4 continued

- (c) (i) Sketch the function $g(x)$ which is defined by $g(x) = \begin{cases} 1 + \frac{|x|}{x}, & \text{for } x \neq 0 \\ 2, & \text{for } x = 0 \end{cases}$. **1**
- (ii) Given $f(x) = -2x$. Sketch $f(g(x))$ on a new number plane. **1**

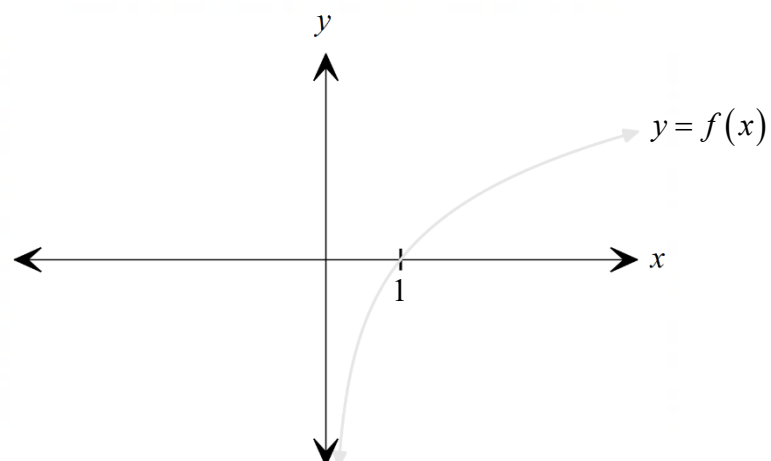
End of Assessment Task

Answer Sheet for Question 1 (b)

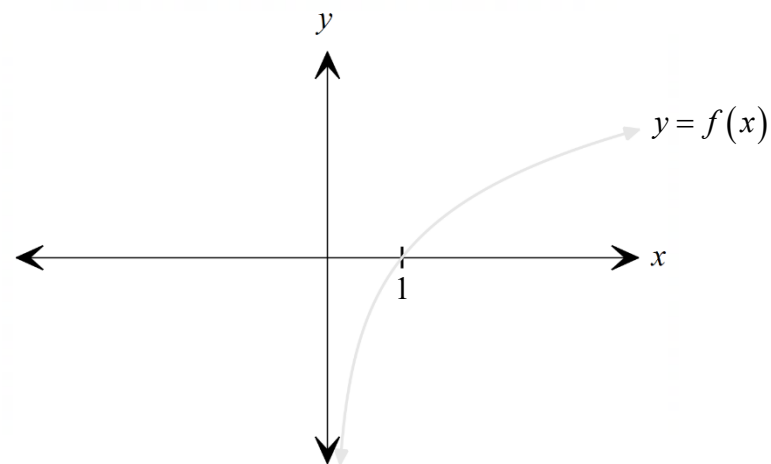
(b) (i) $y = f^{-1}(x)$



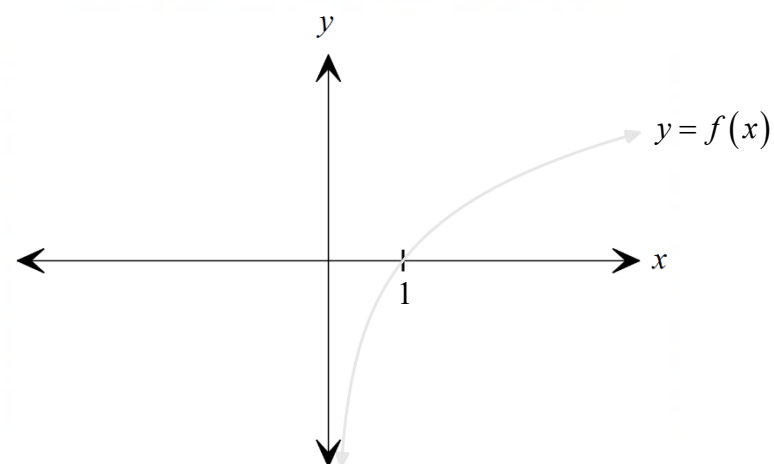
(b) (ii) $y = |f(x)|$



(b) (iii) $y = f(|x|)$



(b) (iv) $y^2 = f(x)$





Suggested Solution (s)

Comments

Question 1

(a) $\frac{1}{2}|f(x)|$ would have reflected curve below x-axis

$\frac{1}{f(x)}$ would have a y-intercept of $\frac{1}{4}$

$(g(x))^2$ would have reflected $\sqrt{f(x)}$ about x-axis.

$\therefore$ B.

(b) see scheme on the next page for requirements each graph.

$$(c) (i) \text{ LHS} = \sqrt{\frac{(1 + \cot^2 x) - \cot^2 x - \cos^2 x}{\cos^2 x}}$$

$$= \sqrt{\frac{1 - \cos^2 x}{\cos^2 x}} \checkmark$$

$$= \sqrt{\frac{\sin^2 x}{\cos^2 x}}$$

$$= \sqrt{\tan^2 x} \checkmark \text{ or } \sqrt{\left(\frac{\sin x}{\cos x}\right)^2}$$

$$= |\tan x|$$

$$(ii) |\tan x| = 1$$

$$\therefore \tan x = \pm 1$$

$$x = 45^\circ, 135^\circ \checkmark \text{ must have both}$$

Question 2

$$(a) \frac{\cos(180^\circ - \theta)}{\cos(90^\circ - \theta)} = \frac{-\cos \theta}{\sin \theta} \\ = -\cot \theta \therefore D.$$

$$(b) \cos(x - 60^\circ) = \frac{1}{\sqrt{2}}$$

$$\text{ref } \angle = 45^\circ$$

$$x - 60^\circ = -45^\circ, 45^\circ$$

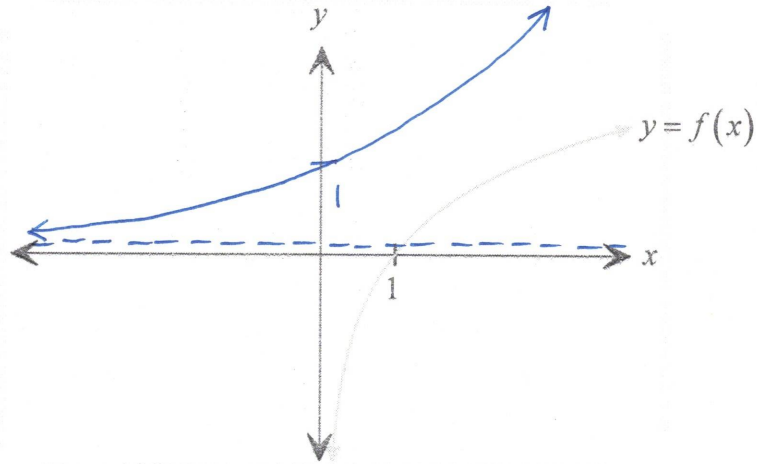
$$x = 15^\circ, 105^\circ \checkmark \text{ both}$$

$$\begin{array}{c|c} S & A \checkmark \\ \hline T & C \checkmark \end{array} \quad -60^\circ \leq x - 60^\circ \leq 300^\circ \checkmark$$

Answer Sheet for Question 1 (b)

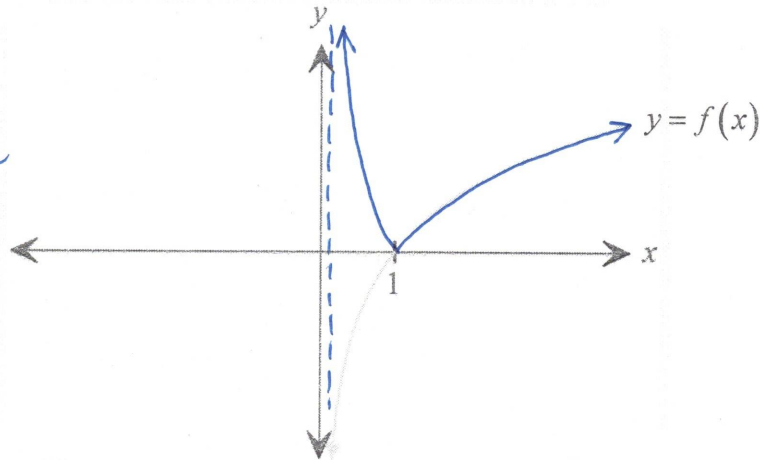
(b) (i) $y = f^{-1}(x)$

- 1 {
- correct shape
 - asymptote
 - y-intercept



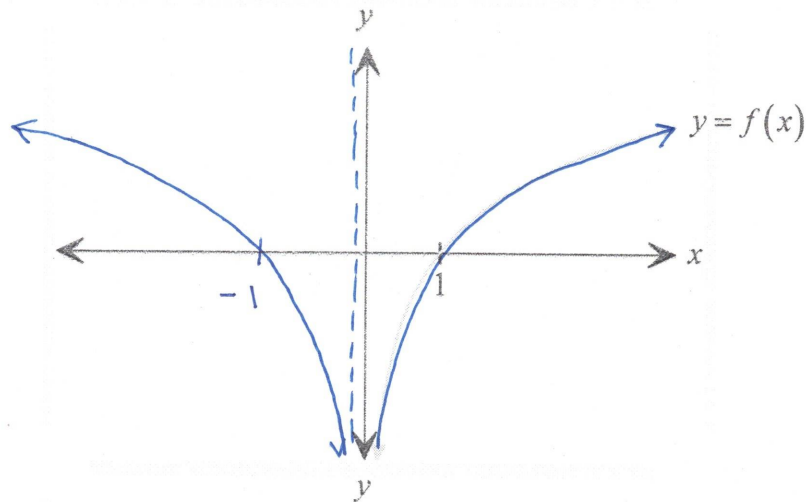
(b) (ii) $y = |f(x)|$

- 1 {
- correct shape
 - asymptote



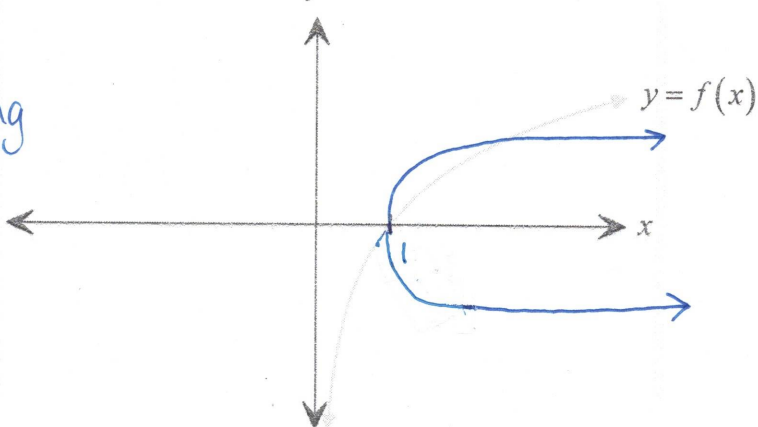
(b) (iii) $y = f(|x|)$

- 1 {
- shape, both sides
 - asymptote
 - x-intercept



(b) (iv) $y^2 = f(x)$

- 1 {
- shape, including cross over
 - reflection in x-axis



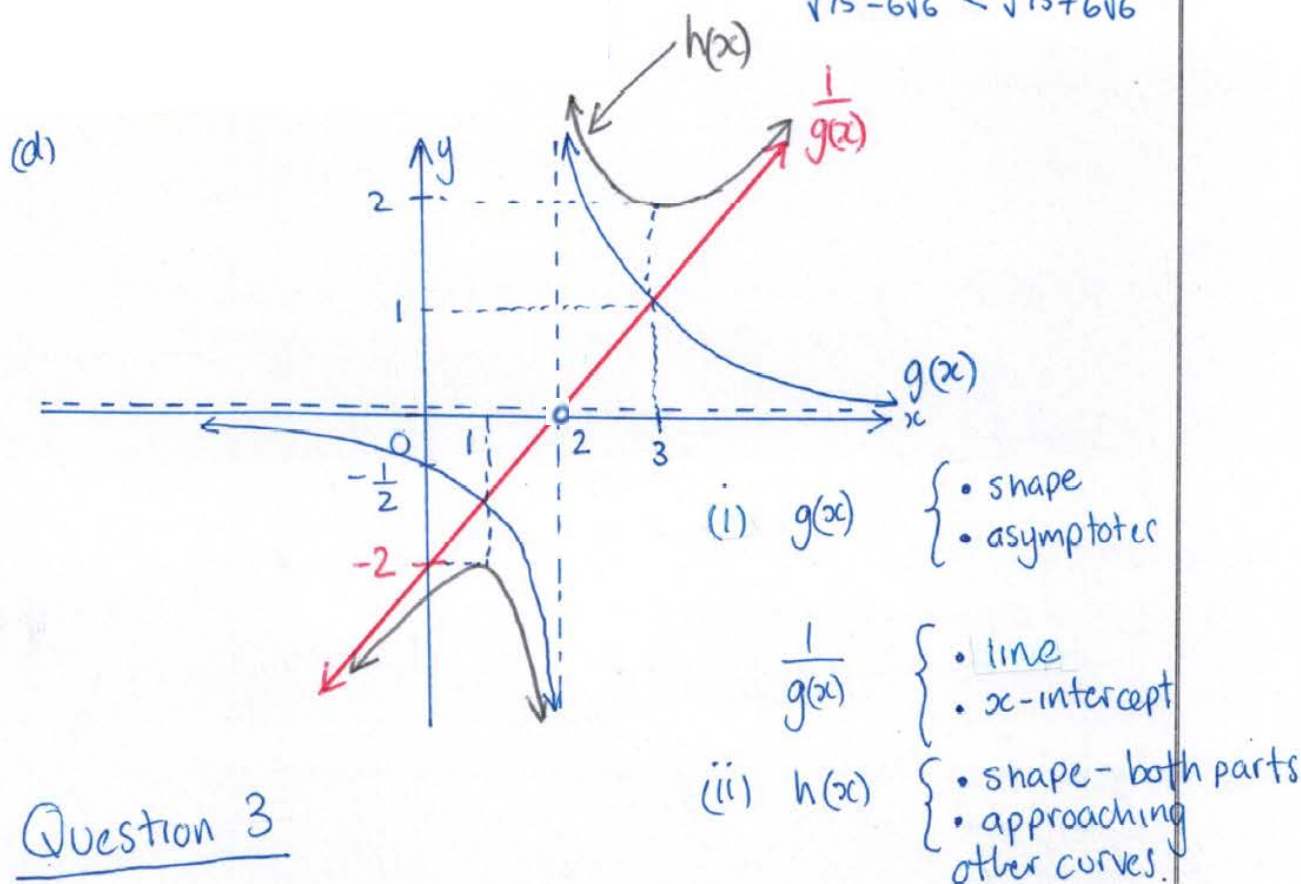


Suggested Solution (s)

Comments

Question 2 (continued)

$$\begin{aligned}
 (c) & \left(\sqrt{15-6\sqrt{6}} \right)^2 - 2 \left(\sqrt{15-6\sqrt{6}} \right) \left(\sqrt{15+6\sqrt{6}} \right) + \left(\sqrt{15+6\sqrt{6}} \right)^2 \\
 &= 15 - 6\sqrt{6} - 2\sqrt{(15^2 - (6\sqrt{6})^2)} + 15 + 6\sqrt{6} \quad \checkmark \\
 &= 30 - 2(3) \\
 &= 24 \quad \therefore \sqrt{15-6\sqrt{6}} - \sqrt{15+6\sqrt{6}} = -\sqrt{24} = -2\sqrt{6} \quad \checkmark \\
 & \quad \quad \quad \sqrt{15-6\sqrt{6}} < \sqrt{15+6\sqrt{6}}
 \end{aligned}$$



Question 3

(a) $R(x) \neq 0 \therefore 0 \text{ or } 1 \Rightarrow C$

(b)(i) $x^3 + bx^2 + cx + d = x(x-3)(x+3)$
 $= x^3 - 9x \quad \checkmark$
 $\therefore \left. \begin{aligned} b &= 0 \\ d &= 0 \\ c &= -9 \end{aligned} \right\} \checkmark$

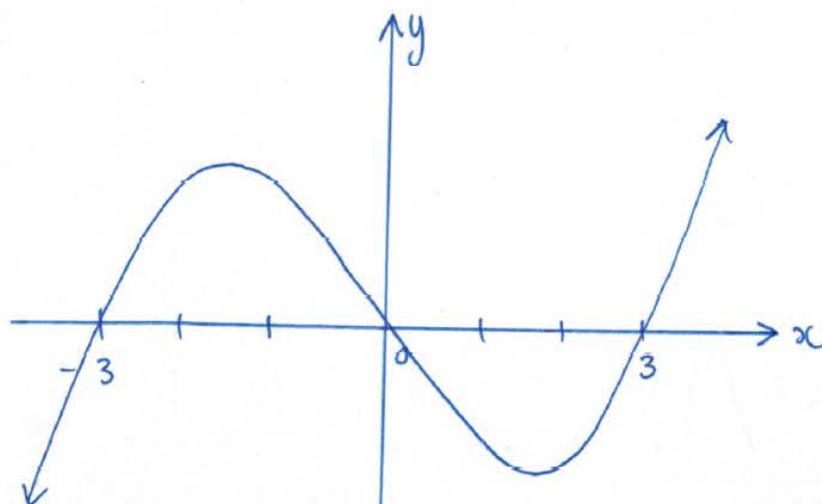


Suggested Solution (s)

Comments

Question 3 (continued)

(b) (ii)



(iii) $x^2 \times \frac{x^2 - 9}{x} > 0$

$$x(x+3)(x-3) > 0$$

$$-3 < x < 0 \text{ or } x > 3$$

$$\text{or } (-3, 0), (3, \infty)$$

(c) $x^3 - 6x^2 - x - 30$

roots are $\alpha, \beta, \alpha + \beta$

sum: $\alpha + \beta + \alpha + \beta = -\frac{6}{1}$ product: $\alpha\beta(\alpha + \beta) = -\frac{30}{1}$

$$2(\alpha + \beta) = -6$$

$$\alpha + \beta = -3$$

$$-3\alpha\beta = 30$$

$$\alpha\beta = -10$$

form a quadratic whose roots have a sum -3
and a product -10

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$x^2 + 3x - 10 = 0$$

$$(x+5)(x-2) = 0$$

roots are $2, -5, -3$ 1 mark
sum and
product1 mark for
utilising line!1 mark
final solution.

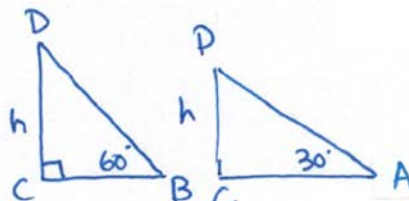


Suggested Solution (s)

Comments

Question 4

(a) B

(b) (i)  $\tan 60^\circ = \frac{h}{CB}$ $\tan 30^\circ = \frac{h}{CA}$ ✓ both equations
 $\sqrt{3} = \frac{h}{CB}$ $\frac{1}{\sqrt{3}} = \frac{h}{CA}$
 $\therefore CB = \frac{h}{\sqrt{3}}, CA = h\sqrt{3}$

(ii) $AB^2 = BC^2 + AC^2 - 2(BC)(AC) \cos 60^\circ$
 $AB^2 = \left(\frac{h}{\sqrt{3}}\right)^2 + (h\sqrt{3})^2 - \frac{h}{\sqrt{3}} \times h\sqrt{3} \quad (\cos 60^\circ = \frac{1}{2})$

$$7^2 = \frac{h^2}{3} + 3h^2 - h^2$$

$$\frac{7}{3}h^2 = 49$$

$$h^2 = 21$$

$$\therefore h = \sqrt{21} \text{ as required.}$$

(iii) $CB = \frac{\sqrt{21}}{\sqrt{3}} = \sqrt{7}$, $CA = \sqrt{21} \times \sqrt{3} = 3\sqrt{7}$ ✓ both.

$$A = \frac{1}{2}ab \sin C$$

$$= \frac{1}{2}(\sqrt{7})(3\sqrt{7}) \sin 60^\circ$$

$$= \frac{21\sqrt{3}}{4} \checkmark$$

(iv) $\frac{7}{2}W = \frac{21\sqrt{3}}{4}$

$$W = \frac{21\sqrt{3}}{4} \times \frac{2}{7}$$

$$W = \frac{3\sqrt{3}}{2} \text{ m } \checkmark$$

one correctly
using cosine
rule

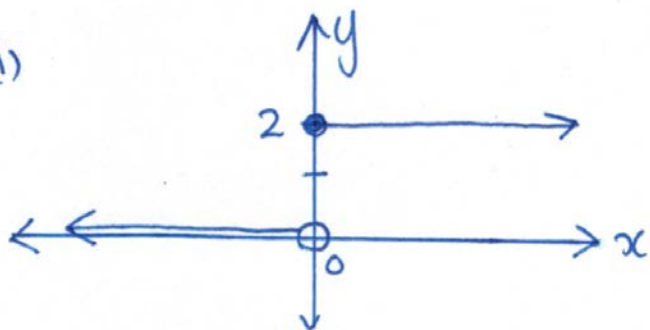


Suggested Solution (s)

Comments

Question 4 (continued)

(c)(i)



$$\begin{aligned} x > 0 & \quad g(x) = 2 \\ x = 0 & \quad g(x) = 2 \\ x < 0 & \quad g(x) = 0 \end{aligned}$$

(ii)

